1 Relatively Prime Numbers, Euler's Totient and $\mathbb{Z}_n^*$

1.1 Relatively prime (co-prime) numbers

Two positive integers a and b are relatively prime (co-prime) if $\gcd(a, b) = 1$, that is, if their only common positive divisor is 1. Neither number has to be prime: the condition is about the pair, not about each number. Since $12 = 2^2 \cdot 3$ and $77 = 7 \cdot 11$ share no prime factor, they are co-prime although both are composite, while $15 = 3 \cdot 5$ and $21 = 3 \cdot 7$ are not, because 3 divides both. If p is prime, every integer from 1 to $p - 1$ is co-prime to p, since the only divisors of p are 1 and p itself.

Cryptography cares about this property for one reason: a **has a multiplicative inverse modulo** n **exactly when** $\gcd(a, n) = 1$. If $ax \equiv 1 \pmod{n}$, then $ax - kn = 1$ for some integer k, so any common divisor of a and n divides 1. Conversely, if $\gcd(a, n) = 1$, the extended Euclidean algorithm (§**??**) produces s, t with $sa + tn = 1$, and reducing modulo n gives $sa \equiv 1$. Co-primality is therefore always checked with the Euclidean algorithm, never by eye.

a, b	Divisors / Euclid steps	GCD	Co-prime?	Note
4, 13	$\{1, 2, 4\}$, $\{1, 13\}$	1	yes	composite + prime
15, 21	$\{1, 3, 5, 15\}$, $\{1, 3, 7, 21\}$	3	**no**	share the factor 3
11, 17		1	yes	both prime
11, 21		1	yes	prime + composite ($21 = 3 \cdot 7$)
12, 77		1	yes	**both composite**
83, 19	$83 = 4 \cdot 19 + 7$, $19 = 2 \cdot 7 + 5$, $7 = 5 + 2$, $5 = 2 \cdot 2 + 1$	1	yes	
790, 121	$790 = 6 \cdot 121 + 64$, $121 = 64 + 57$, $64 = 57 + 7$, $57 = 8 \cdot 7 + 1$	1	yes	lecture homework

1.2 Euler's totient function $\phi(n)$ ★

▸ **Likely this year:** $\phi(n)$ for a given n (Forouzan: "find $\phi(29)$, $\phi(32)$, $\phi(80)$, $\phi(100)$, $\phi(101)$"), and $\phi(n) = (p - 1)(q - 1)$ inside every RSA question.

Many results in public-key cryptography (Euler's theorem, RSA key generation, inverses by exponentiation) need to know *how many* residues modulo n are invertible. Since invertibility is the same as being co-prime to n, this count gets its own name. Euler's totient (phi) function $\phi(n)$ is the number of integers in $\{1, 2, \ldots, n - 1\}$ that are relatively prime to n. Zero and negative numbers are not counted. The lecture videos write it as $\Phi(n)$; Forouzan and the slides write $\phi(n)$.

By hand, $\phi(n)$ is found by computing $\gcd(k, n)$ for every k from 1 to $n - 1$ and counting the 1s. For $n = 8$:

k	1	2	3	4	5	6	7	
$\gcd(k, 8)$	1	2	1	4	1	2	1	
co-prime?	✓	×	✓	×	✓	×	✓	$\Rightarrow \phi(8) = \mathbf{4}$

In the same way $\phi(5) = 4$ and $\phi(11) = 10$, since every number below a prime is co-prime to it. This direct count needs $n - 1$ GCDs, which is useless for $n = 1000$ and hopeless for a 1024-bit RSA modulus, so in practice $\phi(n)$ is always computed from the prime factorisation of n using the rules below.

1.3 Computing $\phi(n)$: the four rules ★

The slides (following Forouzan) give four rules. Each one follows from counting which numbers *fail* to be co-prime to n, and the derivations are short enough to reproduce in an exam.

Rule 1: $\phi(1) = 1$ **by convention** (the slides write $\phi(1) = 0$; see the corrections box). It never matters in practice.

Rule 2: for a prime p**,** $\phi(p) = p - 1$**.** The only divisors of p are 1 and p, so none of $1, \ldots, p - 1$ shares a factor with p. For example $\phi(7) = 6$, $\phi(13) = 12$, $\phi(31) = 30$.

Rule 3: for a prime power, $\phi(p^k) = p^k - p^{k-1}$**.** A number shares a factor with p^k exactly when it is a multiple of p, because p is the only prime in p^k. Among $1, \ldots, p^k$ the multiples of p are $p, 2p, \ldots, p^{k-1} \cdot p$, which is p^{k-1} numbers, and removing them leaves $p^k - p^{k-1} = p^{k-1}(p - 1)$. For example $\phi(8) = 8 - 4 = 4$, $\phi(25) = 25 - 5 = 20$, and $\phi(49) = 49 - 7 = 42$.

Rule 4: if $\gcd(m, n) = 1$**, then** $\phi(mn) = \phi(m)\,\phi(n)$ (the function is *multiplicative*). By the Chinese remainder theorem (§**??**), each x modulo mn corresponds to exactly one pair (x mod m, x mod n), and x is co-prime to mn exactly when it is co-prime to both m and n. The co-prime x therefore correspond to pairs chosen from $\phi(m)$ values and $\phi(n)$ values, giving $\phi(m)\phi(n)$ of them. The condition $\gcd(m, n) = 1$ is essential: $\phi(4) = 2$ but $\phi(2)\phi(2) = 1$.

Two distinct primes: $\phi(pq) = (p - 1)(q - 1)$**.** This is the RSA case, and it is worth deriving directly. Among $1, \ldots, pq$, the numbers sharing a factor with pq are the multiples of p (there are q of them: $p, 2p, \ldots, qp$) and the multiples of q (there are p of them). The number pq itself is in both lists and has been removed twice, so

$$\phi(pq) = pq - q - p + 1 = (p - 1)(q - 1).$$

For example $\phi(21) = 2 \cdot 6 = 12$, $\phi(77) = 6 \cdot 10 = 60$, and $\phi(187) = \phi(11 \cdot 17) = 10 \cdot 16 = 160$.

General formula. Combining rules 3 and 4, if $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$ with distinct primes p_i, then

$$\phi(n) = \left(p_1^{e_1} - p_1^{e_1-1}\right)\left(p_2^{e_2} - p_2^{e_2-1}\right)\cdots\left(p_k^{e_k} - p_k^{e_k-1}\right) = n\left(1 - \tfrac{1}{p_1}\right)\left(1 - \tfrac{1}{p_2}\right)\cdots\left(1 - \tfrac{1}{p_k}\right).$$

The second form works for every n and contains rules 2 and 3 and the pq formula as special cases. Only the *distinct* primes appear in the product; the exponents are already accounted for by the factor n. As a consequence, computing $\phi(n)$ is exactly as hard as factoring n, which is the reason RSA can publish n but must keep $\phi(n)$ secret.

★ MUST KNOW

$$\phi(p) = p - 1, \qquad \phi(p^k) = p^k - p^{k-1}, \qquad \phi(pq) = (p-1)(q-1)\ (p \neq q), \qquad \phi(n) = n \prod_{p \mid n}\left(1 - \tfrac{1}{p}\right).$$

$\phi(mn) = \phi(m)\phi(n)$ **only when** $\gcd(m, n) = 1$. Always factorise n completely first.

Example: $\phi(35)$, with $35 = 5 \times 7$: $(5-1)(7-1) = 4 \times 6 =$ **24**. The grid below checks it by crossing out the multiples of 5 and of 7 among $1, \ldots, 34$; none is a multiple of both, since the first common multiple is 35.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34

red = multiple of 5 (6 numbers), blue = multiple of 7 (4 numbers); 34 − 10 = 24 white cells

Trap: repeated prime factors

$25 = 5 \times 5$ is **not** a product of two distinct primes, so $(5-1)(5-1) = 16$ is wrong. Use rule 3: $\phi(25) = 25 - 5 =$ **20**. (Check: among $1, \ldots, 24$ only 5, 10, 15, 20 share a factor with 25, leaving $24 - 4 = 20$.) The slides make the same point with $\phi(49)$: writing $\phi(49) = \phi(7)\phi(7) = 36$ is wrong because 7 and 7 are not co-prime; the correct value is $49 - 7 = 42$.

WORKED EXAMPLE : $\phi(n)$ from the factorisation (Forouzan / slide style)

Q. Find (a) $\phi(29)$ (b) $\phi(32)$ (c) $\phi(80)$ (d) $\phi(100)$ (e) $\phi(101)$ (f) $\phi(240)$ (g) $\phi(1000)$ (h) $\phi(7000)$.
A. Factorise first, then apply the rules.

n	Factorisation	Working	$\phi(n)$
29	prime	$29 - 1$	**28**
32	2^5	$2^5 - 2^4 = 32 - 16$	**16**
80	$2^4 \cdot 5$	$(2^4 - 2^3)(5-1) = 8 \cdot 4$	**32**
100	$2^2 \cdot 5^2$	$(4-2)(25-5) = 2 \cdot 20$	**40**
101	prime	$101 - 1$	**100**
240	$2^4 \cdot 3 \cdot 5$	$(16-8)(3-1)(5-1) = 8 \cdot 2 \cdot 4$	**64**
1000	$2^3 \cdot 5^3$	$1000 \cdot \frac{1}{2} \cdot \frac{4}{5}$ (or $\phi(8)\phi(125) = 4 \cdot 100$)	**400**
7000	$2^3 \cdot 5^3 \cdot 7$	$7000 \cdot \frac{1}{2} \cdot \frac{4}{5} \cdot \frac{6}{7} = 7000 \cdot \frac{24}{70}$	**2400**

Check: a direct count of the co-prime $k < n$ by script gives the same eight values. Lecture homework: $\phi(369) = \phi(3^2)\phi(41) = 6 \cdot 40 =$ **240** and $\phi(372) = \phi(2^2)\phi(3)\phi(31) = 2 \cdot 2 \cdot 30 =$ **120**.

$\phi(n)$ **is even for every** $n > 2$. If $\gcd(k, n) = 1$ then also $\gcd(n - k, n) = 1$, so the co-prime numbers below n come in pairs $\{k, n-k\}$. The two members of a pair can only coincide if $n = 2k$, but then $\gcd(k, n) = k$, which equals 1 only for $n = 2$. Hence for $n > 2$ the co-prime numbers split into disjoint pairs and their count is even; in $\mathbb{Z}_{14}^*$ the pairs are $\{1, 13\}, \{3, 11\}, \{5, 9\}$.

1.4 The sets $\mathbb{Z}_n$ and $\mathbb{Z}_n^*$

$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ is the set of residues modulo n. Every element has an additive inverse in it ($a + (n - a) \equiv 0$), but not every element has a multiplicative inverse. Since cryptosystems such as the affine cipher, Hill cipher and RSA need to *divide*, they work in the smaller set

$$\mathbb{Z}_n^* = \{\, a \in \mathbb{Z}_n : \gcd(a, n) = 1 \,\},$$

which is exactly the set of residues that have multiplicative inverses. By definition its size is $|\mathbb{Z}_n^*| = \phi(n)$. The slide's rule of thumb: use $\mathbb{Z}_n$ when additive inverses are needed and $\mathbb{Z}_n^*$ when multiplicative inverses are needed. $\mathbb{Z}_n^*$ is closed under multiplication modulo n, because a product of two numbers co-prime to n is again co-prime to n (no prime factor of n can appear in it). Together with the inverses this makes $\langle \mathbb{Z}_n^*, \times \rangle$ a group of order $\phi(n)$ (§**??**), and the theorems of the next section are statements about this group. The slides single out the two sets

with a *prime* modulus, $\mathbb{Z}_p$ and $\mathbb{Z}_p^*$. When $n = p$ is prime, $\mathbb{Z}_p^* = \{1, 2, \ldots, p-1\}$ contains every non-zero residue (e.g. $\mathbb{Z}_{13}^* = \{1, \ldots, 12\}$), so every non-zero element of $\mathbb{Z}_p$ can be divided by, which is what makes $\mathbb{Z}_p$ the field GF(p).

n	$\mathbb{Z}_n^*$	$\lvert\mathbb{Z}_n^*\rvert = \phi(n)$
6	$\{1, 5\}$	$\phi(2)\phi(3) = 2$
7	$\{1, 2, 3, 4, 5, 6\}$	$7 - 1 = 6$
10	$\{1, 3, 7, 9\}$	$\phi(2)\phi(5) = 4$
14	$\{1, 3, 5, 9, 11, 13\}$	$\phi(2)\phi(7) = 1 \cdot 6 = 6$ (slide Example 5)
21	$\{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$	$\phi(3)\phi(7) = 2 \cdot 6 = 12$
26	the 12 odd numbers except 13	$\phi(2)\phi(13) = 12$ (valid keys/determinants mod 26)

1.5 Why cryptography cares about $\phi(n)$

Three later topics rest on $\phi(n)$. First, Euler's theorem (§**??**) states that $a^{\phi(n)} \equiv 1 \pmod{n}$ for every $a \in \mathbb{Z}_n^*$, which allows exponents to be reduced modulo $\phi(n)$ and inverses to be computed as $a^{\phi(n)-1}$. Second, RSA chooses $n = pq$ and computes $\phi(n) = (p-1)(q-1)$; the keys satisfy $ed \equiv 1 \pmod{\phi(n)}$, and Euler's theorem then makes decryption undo encryption. In the standard example $n = 187 = 11 \cdot 17$, $\phi(187) = 160$ and $7 \cdot 23 = 161 \equiv 1 \pmod{160}$, which is why $88^7 \bmod 187 = 11$ and $11^{23} \bmod 187 = 88$ undo each other. Third, the number of primitive roots of a prime p is $\phi(p-1)$, which matters for Diffie–Hellman and ElGamal.

Security. Knowing $\phi(n)$ for an RSA modulus is equivalent to knowing its factors. From $n = pq$ and $\phi(n) = n-(p+q)+1$ one gets $p + q = n - \phi(n) + 1$, and p, q are then the roots of $z^2 - (p+q)z + n = 0$. For $n = 143$ and $\phi(n) = 120$: $p + q = 24$, $pq = 143$, so $p, q =$ **11, 13**. This is why $\phi(n)$ is part of the private key.

Corrections to the notes / slides

- All values recomputed by script: $\phi(5) = 4$, $\phi(8) = 4$, $\phi(11) = 10$, $\phi(35) = 24$, $\phi(25) = 20$, $\phi(49) = 42$, $\phi(240) = 64$, $\phi(1000) = 400$, $\phi(7000) = 2400$, $|\mathbb{Z}_{14}^*| = 6$, $|\mathbb{Z}_{21}^*| = 12$. The homework answers in the old notes ($\phi(369) = 240$, $\phi(372) = 120$) are correct.
- Slide L10-06 gives $\phi(1) = 0$, which follows Forouzan's "integers smaller than n" wording. The standard convention is $\phi(1) = 1$ (count $1 \le k \le n$). It only affects $n = 1$ and never matters in RSA.
- The lecture video's "case 3" ("either or both numbers composite") makes the product formula $n \prod(1 - 1/p)$ look like a last resort. It actually works for every n, and the prime and pq formulas are special cases of it.

Check your understanding

1. **Define $\phi(n)$ and explain why $\phi(p) = p - 1$ for a prime p.** $\phi(n)$ is the number of integers in $1, \ldots, n-1$ co-prime to n. A prime has no divisors other than 1 and itself, so all $p - 1$ numbers below it are co-prime to it.
2. **Why is $\phi(49) \neq \phi(7) \cdot \phi(7)$?** Multiplicativity needs co-prime factors, and $\gcd(7, 7) = 7$. Rule 3 gives $\phi(49) = 49 - 7 = 42$, not 36.
3. **Find $\phi(36)$ and $\phi(60)$.** $36 = 2^2 \cdot 3^2$, so $\phi = 2 \cdot 6 = 12$. $60 = 2^2 \cdot 3 \cdot 5$, so $\phi = 2 \cdot 2 \cdot 4 = 16$.
4. **Derive $\phi(pq) = (p-1)(q-1)$.** Of $1, \ldots, pq$, the q multiples of p and the p multiples of q are not co-prime to pq, and pq is counted in both lists, so $\phi(pq) = pq - p - q + 1 = (p-1)(q-1)$.
5. **List $\mathbb{Z}_{15}^*$ and state its size.** $\{1, 2, 4, 7, 8, 11, 13, 14\}$, which has $\phi(15) = 2 \cdot 4 = 8$ elements.
6. **Why is $\phi(n)$ even for $n > 2$?** The co-prime residues pair up as $k \leftrightarrow n - k$, and the two members of a pair are never equal when $n > 2$.
7. **An RSA modulus is $n = 143$ and someone learns $\phi(n) = 120$. What can they do?** They get $p + q = n - \phi(n) + 1 = 24$ and $pq = 143$, so $p = 11$, $q = 13$; the modulus is broken. Hence $\phi(n)$ must stay secret.
8. **What is the difference between $\mathbb{Z}_n$ and $\mathbb{Z}_n^*$?** $\mathbb{Z}_n$ contains all residues $0, \ldots, n-1$ and is used where addition and additive inverses are needed. $\mathbb{Z}_n^*$ contains only the $\phi(n)$ residues co-prime to n, which are exactly those with multiplicative inverses.